Reader Week 1

Maurice Koster

Overview

Sequences and series are the language of "step-by-step change". They show up whenever we observe a quantity over time (temperatures, prices, interest rates), whenever we run an algorithm that produces successive approximations (root finding, numerical integration, gradient descent), and whenever we model dynamics (difference equations, learning, feedback systems). In all of these settings, the central question is:

> **Do these successive values settle down—and if so, to what?**

That question is the mathematical notion of **convergence**. Once you can decide whether a sequence converges, you can reason reliably about limits, stability, approximation quality, and long-run behavior.

A **series** is what you get when you start *adding up* the terms of a sequence. Series occur naturally in economics and finance (discounted cash flows), in probability (expected values and tail bounds), and throughout applied mathematics via **power series** (which define familiar functions such as e^x, $\sin x$, and $\cos x$). Here, the key question becomes:

> **Do the partial sums approach a finite value?**

To answer that, we develop practical convergence tools: monotonicity and boundedness, alternating series arguments, absolute convergence, and the ratio test—culminating in a first look at power series and radii of convergence.

Road map

This is an interactive reader. There are also Python programming assignments. Many thanks to the quarto-pyodide team for creating this interactive and easy to use tool.

Part I

Sequences

1 Motivation and basic definitions

Sequences are basic mathematical constructs over the natural numbers, and indispensable in numerous areas of science. They are not just a mathematical concept: they are an integral tool in the realm of science. In fact, examples of sequences and series abound in real-life scenarios, illustrating their practical relevance.

Consider the simple act of measuring the temperature in your garden on a daily basis. You are then actually generating a sequence (a time series) of temperature readings.

You will also find numerous examples of sequences in the world of financial econometrics. Analysts study time series comprised of valuations of financial products such as stocks and bonds. These sequences of financial data unveil crucial information about market trends, volatility, and investment opportunities. By scrutinizing these series, analysts can make informed decisions and devise effective investment strategies.

In the course *Programming and Numerical Analysis*, you were introduced to algorithms that generate sequences of values. These algorithms serve a crucial purpose, because they help estimate fundamental mathematical quantities like derivatives and roots of equations.

So these examples already illustrate that a basic understanding of the theory of sequences contributes to understanding sequential data in a general sense, and to exploring techniques to model and analyze temporal patterns. Let us start more formally.

What is a sequence?

The basic characteristic of all sequences is that one can number their elements. This means that a sequence in a set V is basically a mapping

$$f : \mathbb{N} \to V,$$

possibly after re-numbering and re-labelling.[1]

A sequence is a mathematical structure that can be represented in this way. The image $f(n) \in V$ is called the n-th element, with index n. It will be convenient to denote the n-th element of the sequence as

$$a_n := f(n).$$

[1] Here we use the convention that $\mathbb{N} = \{1, 2, ...\}$; sometimes it will be useful to extend the set $\mathbb{N}$ with the 0 element: $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$.

A sequence can then also be written as $\{a_n\}_{n\in\mathbb{N}}$.

Other times it will be more convenient to use other sub-indices. For instance you may want to define a sequence through

$$a_n = \frac{1}{n-1},$$

but clearly this will function only if the sequence of indices starts at $n = 2$. In this case you could transform your sequence by relabelling: introduce the sequence $\{b_n\}_{n\in\mathbb{N}}$ defined by $b_n := a_{n+1}$ for $n \in \mathbb{N}$. This sequence would agree with the mapping $f(n) = \frac{1}{n}$.

Re-labelling is not really necessary: you could start numbering at $n = 2$, so that the right expression becomes $\{a_n\}_{n=2}^{\infty}$.[2]

Example 1.1.

- A trivial example of a sequence in $\mathbb{N}$ is the mapping $f(n) = n$.
- The sequence of all even numbers can be associated with the mapping $f(n) = 2n$.
- The sequence of all odd numbers can be associated with the mapping $f(n) = 2n + 1$.

Numerous other examples of sequences will be given in the subsequent chapters. Below we will first focus on sequences in $\mathbb{R}$: convergence, monotonicity, and the Monotonic Convergence Theorem. Then we extend convergence to sequences in higher-dimensional spaces $\mathbb{R}^m$ using norms and distances.

[2] This is the reason that sequences are sometimes denoted $\{a_n\}$ if there is common ground about the indices. In this case we could have known that $\{a_n\} = \{a_n\}_{n=2}^{\infty}$. For example, this approach is used in Stewart, Chapter 11 on sequences and series.

2 Sequences in $\mathbb{R}$ and convergence

2.1 Definition of convergence

We say that the sequence $\{a_n\}_{n\in\mathbb{N}}$ in $\mathbb{R}$ has limit $L \in \mathbb{R}$ if, for large enough indices, the elements become arbitrarily close to L.

Definition 2.1 (Convergence of sequences in $\mathbb{R}$)**.** The sequence $\{a_n\}_{n\in\mathbb{N}}$ has limit $L \in \mathbb{R}$ if for each $\varepsilon > 0$ there exists an index number $N \in \mathbb{N}$ such that

$$n \geq N \Rightarrow |a_n - L| \leq \varepsilon.$$

Shorthand notation:

$$\lim_{n\to\infty} a_n = L.$$

If a sequence does not converge, it is called *divergent*.

Example 2.1 (Random sequence)**.** Generate a sequence $\{a_n\}_{n=1}^{\infty}$ using

$$a_n = 1 + \frac{(-1)^n + \delta}{n},$$

where δ is randomly drawn from the uniform distribution on $[-1, 1]$. Then $\lim_{n\to\infty} a_n = 1 = L$.

Let $\varepsilon > 0$ be arbitrary but fixed. For all n we have

$$-\tfrac{2}{n} \leq 1 - a_n \leq \tfrac{2}{n} \quad \Rightarrow \quad |a_n - L| \leq \tfrac{2}{n}.$$

If $\frac{2}{n} \leq \varepsilon$ then $|a_n - L| \leq \varepsilon$, which is guaranteed when $n \geq N := \lceil \frac{2}{\varepsilon} \rceil$.[1]

Example 2.2 (Geometric sequence)**.** Consider the geometric sequence $a_n = r^n$ for $n \in \mathbb{N}$, and assume $r \neq 0$. Then a_n converges for $-1 < r \leq 1$.

If $r = 1$ convergence is clear, as $a_n = 1$ for all n. If $r = -1$, divergence is also clear as $a_n = (-1)^n$.

[1]Here $\lceil\cdot\rceil$ denotes the ceiling function, $\lceil x \rceil := \min\{k \in \mathbb{Z} : k \geq x\}$.

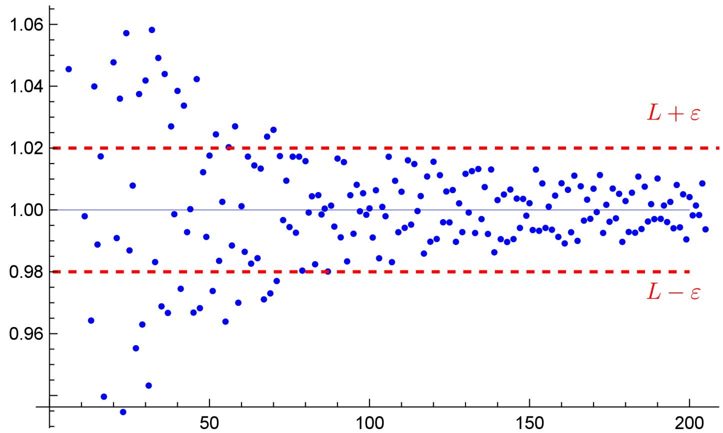

Figure 2.1: Illustration of the limit in Example 1.

Now suppose $|r| < 1$. We will show that the limit is $L = 0$. Take $\varepsilon > 0$, arbitrary but fixed. Then

$$\begin{aligned} |a_n - L| = |a_n| \leq \varepsilon \Leftrightarrow |r|^n &\leq \varepsilon \\ \Leftrightarrow n \ln(|r|) &= \ln(|r|^n) \leq \ln(\varepsilon) \\ \Leftrightarrow n &\geq \frac{\ln(\varepsilon)}{\ln(|r|)}. \end{aligned}$$

The last inequality uses $\ln(|r|) < 0$. So let

$$N := \left\lceil \frac{\ln(\varepsilon)}{\ln(|r|)} \right\rceil,$$

then $n \geq N \Rightarrow |a_n - L| \leq \varepsilon$.

Geometric sequences play an important role in finance, as these can be used to model processes with a constant proportional change (e.g., steady growth, inflation, or discount factors), where each period's value is obtained by multiplying the previous one by the same factor.

Example 2.3 (Arithmetic sequence)**.** Consider the arithmetic sequence inductively defined, i.e. $a_1 = A, a_n = a_{n-1} + K$ for $n \in \mathbb{N}$. This sequence clearly does not converge, or it must be constant.

Arithmetic sequences also appear in finance. Whenever a cash flow, payment, or value changes by a constant amount per period—such as in stepwise contributions, fixed annual raises in income, or straight-line depreciation—the underlying quantities follow an arithmetic pattern.

Exercises

Exercise 2.1. Show that $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$.

As seen in Exercise 2.1, the formula can be proved by induction.

3 Monotonic convergence

In this chapter we examine two qualitative properties that are very useful to show convergence: **boundedness** and **monotonicity**. Many times we do not have a closed formula for the sequence (or it is hard to obtain), but these qualitative tools still let us prove convergence.

Definition 3.1 (Monotonicity)**.** A sequence $\{a_n\}_{n\in\mathbb{N}}$ is *monotonic* if either:

- $a_n \geq a_{n+1}$ for all n (monotonically decreasing), or
- $a_{n+1} \geq a_n$ for all n (monotonically increasing).

Definition 3.2 (Boundedness)**.** A sequence $\{a_n\}_{n\in\mathbb{N}}$ is *bounded* if there is a constant $M > 0$ such that

$$-M \leq a_n \leq M \quad \text{for all } n.$$

Theorem 3.1 (Monotonic Convergence Theorem)**.** *Suppose $\{a_n\}_{n\in\mathbb{N}}$ is bounded and monotonic. Then the limit*

$$L = \lim_{n\to\infty} a_n$$

exists.

Proof (increasing case)

Assume $\{a_n\}$ is monotonically increasing and bounded. Then

$$L = \sup\{a_n \mid n \in \mathbb{N}\}$$

exists. Take $\varepsilon > 0$. Since L is the least upper bound, $L - \varepsilon$ is not an upper bound, so there exists N with $L - \varepsilon < a_N \leq L$. Because the sequence is increasing, for all $n \geq N$ we have $L - \varepsilon < a_N \leq a_n \leq L$, hence $|a_n - L| < \varepsilon$.
A proof for decreasing sequences is similar (use the infimum).

Example 3.1 (An iteratively defined sequence)**.** Consider the sequence defined by

$$a_1 = 1, \qquad a_{n+1} = 3 - \frac{1}{a_n}.$$

We want to understand the long-term behavior and compute $\lim_{n\to\infty} a_n$.

We show: - $\{a_n\}$ is increasing, - $\{a_n\}$ is bounded.

Increasing (induction). Define $P(n) : a_{n+1} > a_n$. Then $P(1)$ holds since $a_2 = 3 - 1/a_1 = 2 > a_1$. Assume $P(k)$ holds so that $a_{k+1} > a_k > \cdots > a_1 = 1$. Then

$$a_{k+2} = 3 - \frac{1}{a_{k+1}} > 3 - \frac{1}{a_k} = a_{k+1},$$

so $P(k+1)$ holds.

Bounded. One can show $a_n \leq 3$ for all n (and all terms stay positive).

By the *Monotonic Convergence Theorem*, $L = \lim_{n\to\infty} a_n$ exists. Compute:

$$L = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty}\left(3 - \frac{1}{a_n}\right) = 3 - \frac{1}{L}.$$

So L solves $L = 3 - \frac{1}{L}$, i.e.

$$L = \tfrac{3}{2} + \tfrac{1}{2}\sqrt{5}$$

(the other root is not compatible with the sequence).

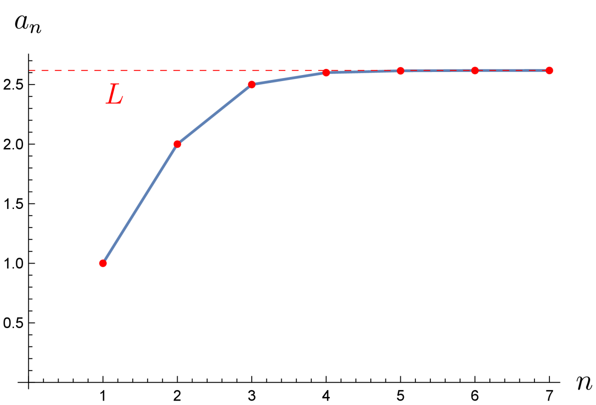

Figure 3.1: Sequence values and (fast) convergence to L.

4 Sequences and continuity

Continuity

Using functions $f : \mathbb{R} \to \mathbb{R}$ it is easy to generate a new sequence $\{b_n\}_{n\in\mathbb{N}}$ from a given sequence $\{a_n\}_{n\in\mathbb{N}}$ by applying f to each element:

$$b_n := f(a_n).$$

Below we focus on continuous f. First we will repeat what that actually means.

Definition 4.1 (Continuity)**.** Consider a function $f : I \to \mathbb{R}$, where $I \subset \mathbb{R}$ is an open interval, and take $a \in I$. Then f is *continuous at* a if

$$\lim_{x\to a} f(x) = f(a).$$

Equivalently: for each $\varepsilon > 0$ there is $\delta > 0$ such that

$$x \in I,\ |x - a| \leq \delta \Rightarrow |f(x) - f(a)| \leq \varepsilon.$$

The function f is *continuous on* I if it is continuous at every $a \in I$.

Continuity and row continuity

Theorem 4.1. ∼ *Characterization of continuity in terms of sequences*

*Let $f : I \to \mathbb{R}$ with $I \subset \mathbb{R}$ open. Then f is continuous at $a \in I$ **iff** for each sequence $\{a_n\} \subset I$ with $a_n \to a$ we have*

$$f(a_n) \to f(a).$$

Proof

(⇒) If f is continuous at a, then for each $\varepsilon > 0$ there exists $\delta > 0$ such that $|x - a| \leq \delta \Rightarrow |f(x) - f(a)| \leq \varepsilon$. If $a_n \to a$, then for some N we have $n \geq N \Rightarrow |a_n - a| \leq \delta$, hence $|f(a_n) - f(a)| \leq \varepsilon$.

($\Leftarrow$) Suppose the sequential condition holds but f is not continuous at a. Then there exists $\varepsilon > 0$ such that for every n there exists $b_n \in (a - \frac{1}{n}, a + \frac{1}{n})$ with $|f(b_n) - f(a)| > \varepsilon$. Then $b_n \to a$ but $f(b_n) \not\to f(a)$, contradiction.

A standard consequence: if $a_n \to a$ and f is continuous at a, then $f(a_n) \to f(a)$. For example:

$$\lim_{n\to\infty} \sqrt{1 + \tfrac{1}{n}} = \sqrt{\lim_{n\to\infty} (1 + \tfrac{1}{n})} = \sqrt{1+0} = 1.$$

5 Sequences in $\mathbb{R}^m$

A sequence $\{a_n\}_{n\in\mathbb{N}}$ in $\mathbb{R}^m$ consists of elements $a_n \in \mathbb{R}^m$. So instead of considering numbers only, we now look at m-tuples of numbers.

Example 5.1. Define the sequence $\{a_n\}_{n\in\mathbb{N}}$ in $\mathbb{R}^2$ by

$$a_n := (\cos(n), \sin(n)).$$

The first 200 elements of this sequence are plotted in Figure Figure 5.1.

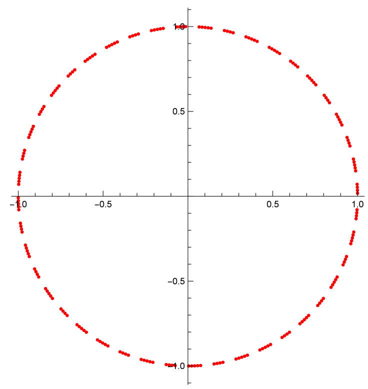

Figure 5.1: Plot of $a_n = (\cos(n), \sin(n))$ for $n = 1, 2, \dots, 200$.

Similar to the discussion of convergence of a sequence in $\mathbb{R}$ one may ask oneself when a sequence in $\mathbb{R}^m$ converges – or what that should mean. Note that $\lim\limits_{n\to\infty} a_n = L$ in $\mathbb{R}$ means that by taking high enough index numbers n, the corresponding sequence elements a_n get arbitrarily close to the value L. Intuitively, we would want to say that a converging sequence $\{a_n\}_{n\in\mathbb{N}} \in \mathbb{R}^m$ must get close to some element of $\mathbb{R}^m$, where the **closeness** is measured by some idea of ***distance***. A distance measure that is quite common is the Euclidean distance $d()$, which uses the Euclidean norm on $\mathbb{R}^m$.

5.1 Norm and distance

A sequence $\{a_n\}_{n\in\mathbb{N}}$ in $\mathbb{R}^m$ consists of elements $a_n \in \mathbb{R}^m$, so instead of numbers we now consider m-tuples of numbers.

A common choice of distance in $\mathbb{R}^m$ is based on the Euclidean norm

$$\|x\| := \sqrt{\sum_{k=1}^{m} x_k^2}.$$

The associated Euclidean distance is

$$d(x, y) = \|x - y\| = \sqrt{\sum_{k=1}^{m} (x_k - y_k)^2}.$$

Properties of norms, general definition

A function $\|\cdot\| : \mathbb{R}^m \to \mathbb{R}$ is a **norm** if for all $x, y \in \mathbb{R}^m$ and $\alpha \in \mathbb{R}$:

1. **Non-negativity:** $\|x\| \geq 0$.
2. **Definiteness:** $\|x\| = 0 \iff x = 0$.
3. **Homogeneity:** $\|\alpha x\| = |\alpha| \, \|x\|$.
4. **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$.

There are different norms on $\mathbb{R}^m$ that have clear applications. Here are some examples.

1. The L_1 norm is defined by $\|x\|_1 := \sum_{i=1}^{m} |x_i|$.
2. The L_∞ norm is defined by $\|x\|_\infty := \max_{i=1,\ldots,m} |x_i|$

What do your think an alternative name is for the Euclidean norm?

Convergence in $\mathbb{R}^m$

The notion of converging sequences in $\mathbb{R}$ is easily extended to sequences in $\mathbb{R}^m$. Now, a converging sequence in $\mathbb{R}^m$ *closes in* on some vector $L \in \mathbb{R}^m$. This means that by picking high enough indices, the distance between the corresponding row elements and L become arbitrary small. We will measure the distance between vectors by the *Euclidean norm* $\| \cdot \|$. Formally,

Definition 5.1 (Convergence in $\mathbb{R}^m$)**.** A sequence $\{a_n\}_{n \in \mathbb{N}} \subset \mathbb{R}^m$ *converges* to $L \in \mathbb{R}^m$ if for each $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n \geq N \Rightarrow \|a_n - L\| \leq \varepsilon.$$

Example 5.2. Define the sequence $\{a_n\}_{n \in \mathbb{N}}$ in $\mathbb{R}^2$ by

$$a_n := \left(1 + \tfrac{1}{n}, \tfrac{1}{n}\right).$$

Then $a_n \to (1, 0)$, because

$$\|a_n - (1, 0)\| = \sqrt{(\tfrac{1}{n})^2 + (\tfrac{1}{n})^2} = \tfrac{\sqrt{2}}{n} \to 0.$$

Example 5.3. Define the sequence $\{a_n\}_{n\in\mathbb{N}}$ in $\mathbb{R}^2$ by

$$a_n := \left(1 + \tfrac{1}{n}, n\right).$$

Then we see that the sequence of the first coordinates $\{a_{n,1}\}_{n\in\mathbb{N}} = \{1 + \frac{1}{n}\}_{n\in\mathbb{N}}$ converges with limit 1, whereas the sequence of second coordinates ({{a_{n,2}}}_{{n ℕ}}) does not converge as $a_{n,2} \to \infty$ for $n \to \infty$. The sequence $\{a_n\}$ does not get close to a particular element of $\mathbb{R}^2$ for $n \to \infty$, so we would intuitively say that it does not converge. In fact, for any fixed $L = (L_1, L_2) \in \mathbb{R}^2$,

$$\|a_n - L\| = \sqrt{(1 - L_1 + \tfrac{1}{n})^2 + (n - L_2)^2} \to \infty.$$

A nice property of norms on $\mathbb{R}^m$

A common property of all norms on $\mathbb{R}^m$ is that they have the same convergence properties. In the definition above you may exchange $\|\cdot\|$ for any norm. Interested? Figure out, or use the web.

Coordinate-wise criterion

In our previous example, what went wrong with respect to convergence is that in one coordinate the values of the vectors did not converge. It is easily seen that a sequence of vectors can only converge only if all coordinates of these vectors converge. It will also be sufficient. We have the following:

Theorem 5.1 (Coordinate-wise convergence criterion)**.** *The sequence $\{a_n\} \subset \mathbb{R}^m$ converges to L **iff** for $k = 1, 2, \dots, m$,*

$$\lim_{n\to\infty} a_{n,k} = L_k.$$

Proof

We show both implications.
(⇒) Suppose $\{a_n\}_{n\in\mathbb{N}} \subset \mathbb{R}^m$ converges to $L \in \mathbb{R}^m$. Let $\varepsilon > 0$. Then there exists N such that for all $n \geq N$,

$$\|a_n - L\| \leq \varepsilon.$$

Write $a_n = (a_{n,1}, \dots, a_{n,m})$ and $L = (L_1, \dots, L_m)$. For each coordinate $k \in \{1, \dots, m\}$ and $n \geq N$,

$$|a_{n,k} - L_k| = \sqrt{(a_{n,k} - L_k)^2} \leq \sqrt{\sum_{j=1}^{m} (a_{n,j} - L_j)^2} = \|a_n - L\| \leq \varepsilon.$$

Hence $\lim_{n\to\infty} a_{n,k} = L_k$ for every k.

(⇐) Conversely, suppose $\lim_{n\to\infty} a_{n,k} = L_k$ for all $k \in \{1, \dots, m\}$. Let $\varepsilon > 0$. For each k there exists N_k such that for all $n \geq N_k$,

$$|a_{n,k} - L_k| \leq \frac{\varepsilon}{\sqrt{m}}.$$

Let $N = \max\{N_1, \dots, N_m\}$. Then for all $n \geq N$,

$$\|a_n - L\| = \sqrt{\sum_{k=1}^{m} (a_{n,k} - L_k)^2} \leq \sqrt{\sum_{k=1}^{m} \frac{\varepsilon^2}{m}} = \varepsilon.$$

Therefore $a_n \to L$ in $\mathbb{R}^m$.

Subsequences

Given a sequence $\{a_n\}_{n\in\mathbb{N}}$ and an increasing function $f : \mathbb{N} \to \mathbb{N}$, the sequence $\{a_{f(n)}\}_{n\in\mathbb{N}}$ is called a *subsequence* of $\{a_n\}_{n\in\mathbb{N}}$.

A subsequence is therefore a sequence which is "drawn" from a sequence of elements.

Example 5.4. Let $\{a_n\}_{n\in\mathbb{N}} = \{(-1)^n\}_{n\in\mathbb{N}}$. Then $f : \mathbb{N} \to \mathbb{N}$ with $f(n) = 2n$ defines the constant subsequence

$$b_n = a_{f(n)} = 1.$$

Theorem of Bolzano–Weierstrass

The following theorem will be important in numerous applications in proofs in the second part of the course. You will also encounter it in the second reader, only then confined to a slightly different setting.

Theorem 5.2 (Bolzano–Weierstrass)**.** *Let $\{x_n\}$ be a bounded sequence in $\mathbb{R}^m$. Then it has a convergent subsequence.*

Proof

First look at the case $m = 1$. Take a bounded sequence $\{x_n\}$ in $\mathbb{R}$. Then there is $M > 0$ with $x_n \in [a_0, b_0] := [-M, M]$ for all n.
Subdivide $[a_0, b_0]$ into two subintervals of equal length. One of these contains infinitely many points of the sequence. Subdivide that subinterval again into two equal parts, and continue. This yields nested intervals

$$[a_1, b_1] \supset [a_2, b_2] \supset \cdots$$

each containing infinitely many points of $\{x_n\}$, with lengths

$$b_k - a_k = \frac{M}{2^{k-1}}.$$

The endpoints $\{a_k\}$ form a non-decreasing sequence bounded above, and $\{b_k\}$ form a non-increasing sequence bounded below, so both converge. Since $b_k - a_k \to 0$, they converge to the same limit ℓ.
Now pick $x_{n_1} \in [a_1, b_1]$, then choose x_{n_2} with $n_2 > n_1$ from $[a_2, b_2]$, and so on. Then for every k,

$$a_k \le x_{n_k} \le b_k,$$

so by the squeeze theorem $x_{n_k} \to \ell$. Hence every bounded sequence in $\mathbb{R}$ has a convergent subsequence.
For general m, write $x_n = (x_{n,1}, \dots, x_{n,m})$. Boundedness of $\|x_n\|$ implies each coordinate sequence $\{x_{n,k}\}$ is bounded. Apply the $m = 1$ result to the first coordinate to obtain a convergent subsequence; then refine it to a subsequence whose second coordinate converges; and so on. After m steps we obtain a subsequence $\{x_{n_k}\}$ for which all components converge, hence x_{n_k} converges in $\mathbb{R}^m$.

Example 5.5. The sequence $\{(-1)^n\}_{n\in\mathbb{N}}$ does not converge, but it is bounded. One convergent subsequence is obtained by $f(n) = 2n$, giving $b_n = a_{2n} = 1$.

Example 5.6. Consider the sequence $a_n = (\cos n, \sin n)$ in $\mathbb{R}^2$. This sequence does not converge. But since $\|a_n\| = 1$ for all n, it is bounded, so by Bolzano–Weierstrass it has a convergent subsequence.

Example 5.7. Let $a_n = (-1)^n$. If we choose the increasing index map $f(n) = 2n$, then

$$b_n := a_{f(n)} = a_{2n} = 1 \quad \text{for all } n,$$

so $\{b_n\}$ is a constant (and hence convergent) subsequence.

Part II

Series

6 Series

Series are constructed using sequences. Consider a sequence $\{a_n\}_{n\in\mathbb{N}}$. We can form the sequence of **partial sums**

$$s_n = \sum_{k=1}^{n} a_k.$$

We call the sequence of partial sums $\{s_n\}_{n\in\mathbb{N}}$ a **series** (associated to $\{a_n\}$). If

$$S = \lim_{n\to\infty} s_n$$

exists, then the series **converges** and S is called its **sum**.

Example 6.1 (Geometric series)**.** Let $r \in \mathbb{R}$ and consider $\{r^n\}_{n=0}^{\infty}$. The partial sums satisfy

$$s_n = \sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r} \quad (r \neq 1).$$

Hence

$$\lim_{n\to\infty} s_n = \begin{cases} \frac{1}{1-r}, & |r| < 1, \\ \text{does not exist}, & |r| \geq 1. \end{cases}$$

Example 6.2 (Harmonic series)**.** Consider the **harmonic series**

$$s_n = \sum_{k=1}^{n} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}.$$

The sequence of partial sums $\{s_n\}_{n=1}^{\infty}$ is increasing. Moreover,

$$
\begin{aligned}
s_2 &= 1 + \frac{1}{2}, \\
s_4 &= 1 + \frac{1}{2} + \overbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}^{>\frac{1}{2}} \;>\; 1 + 2 \cdot \frac{1}{2}, \\
s_8 &= 1 + \frac{1}{2} + \overbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}^{>\frac{1}{2}} + \overbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)}^{>\frac{1}{2}} \;>\; 1 + 3 \cdot \frac{1}{2}, \\
&\vdots \\
s_{2^k} &= 1 + \frac{1}{2} + \overbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}^{>\frac{1}{2}} + \overbrace{\left(\frac{1}{5} + \cdots + \frac{1}{8}\right)}^{>\frac{1}{2}} + \cdots + \overbrace{\left(\frac{1}{2^{k-1}+1} + \cdots + \frac{1}{2^k}\right)}^{>\frac{1}{2}} \;>\; 1 + k \cdot \frac{1}{2}.
\end{aligned}
$$

Hence,

$$\lim_{k\to\infty} s_{2^k} = \infty,$$

so also $\lim_{n\to\infty} s_n = \infty$. Therefore the harmonic series does *not* converge, even though the terms $\frac{1}{k}$ tend to 0.

```
# Calculate the smallest number n such that s_n>=5

print("s_n =", n)
```

We will see below that $a_n \to 0$ is a *necessary* condition for $\sum_{n=1}^{\infty} a_n$ to converge, but it is *not sufficient.* $\triangleleft$

7 Series and convergence

A necessary condition for convergence of a series is that the underlying sequence must have a limit 0.

Theorem 7.1 (Necessary condition for convergence)**.** *If* $\sum_{n=0}^{\infty} a_n$ *converges, then* $\lim_{n\to\infty} a_n = 0$*.*

Proof

Let $s_n = \sum_{k=0}^{n} a_k$. If $s_n \to S$, then also $s_{n+1} \to S$. Hence

$$a_{n+1} = s_{n+1} - s_n \longrightarrow S - S = 0.$$

Do you think it is also a sufficient condition for convergence? Further basic properties of series are these algebraic ones. You may try to prove them yourself:

Theorem 7.2 (Linearity of convergent series)**.** *Let* $\sum_{n=0}^{\infty} a_n$ *and* $\sum_{n=0}^{\infty} b_n$ *converge with sums* S_a *and* S_b*. Then for any* $c \in \mathbb{R}$*:*

1. $\sum_{n=0}^{\infty} (ca_n)$ *converges and has sum* cS_a*;*
2. $\sum_{n=0}^{\infty} (a_n \pm b_n)$ *converges and has sum* $S_a \pm S_b$*.*

8 Alternating series

An *alternating series* has the form

$$\sum_{n=0}^{\infty}(-1)^n a_n,$$

where typically $a_n \geq 0$.

Some examples:

- For $a_n = n$ we have $\sum_{n=0}^{\infty}(-1)^n a_n = \sum_{n=0}^{\infty}(-1)^n n$.
- For $a_n = \frac{1}{n}$ we get the series $\sum_{n=0}^{\infty}(-1)^n \frac{1}{n}$.

It is easy to see that the first example of an alternating series is not converging, since $\lim_{n\to\infty}(-1)^n a_n$ is not convergent, it is divergent. Now what about the second example? See the figure below.

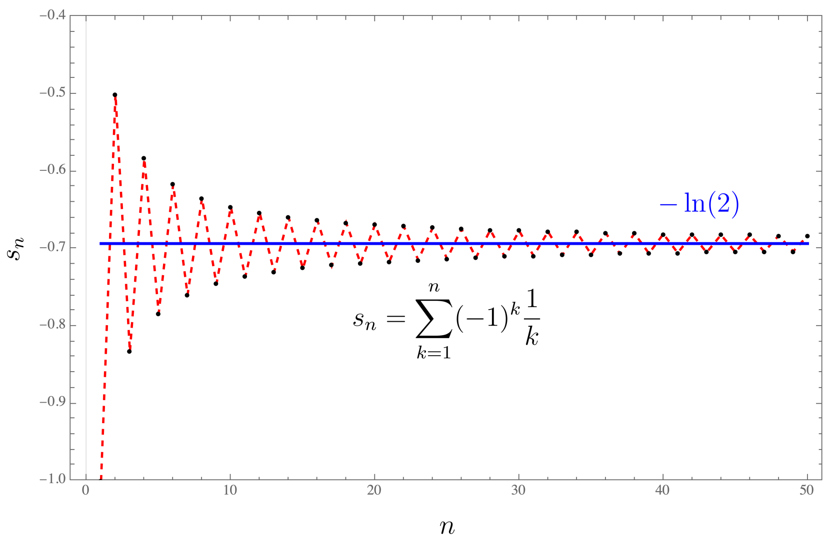

Figure 8.1: Partial sums s_n plotted.

Theorem 8.1 (Alternating Series Test + error bound)**.** *Let $a_n \geq 0$ be decreasing and $a_n \to 0$. Then $\sum_{n=0}^{\infty}(-1)^n a_n$ converges. If S is its sum and s_n the n-th partial sum, then*

$$|S - s_n| \leq a_{n+1}.$$

Proof

Let $b_n = s_{2n}$ and $c_n = s_{2n+1}$. Then for all n we have that $c_n \leq b_n$. Moreover $\{b_n\}$ is decreasing and the elements are bounded below by c_1. Also, $\{c_n\}$ is increasing and the elements are bounded above by b_1. This means that by the Monotonic Convergence Theorem these sequences are both converging. Also $b_n - c_n = a_{2n+1} \to 0$, so both subsequences have the same limit, hence $S_n \to S$.
Moreover, $|S - s_n| \leq |s_{n+1} - s_n| = a_{n+1}$.

Example 8.1. Consider the alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}.$$

This is an alternating series, with $a_n = \frac{1}{n} \downarrow 0$. So the series converges.

In orde to estimate its limit (which is actually $\ln(2)$ as we will see later), we may concentrate on a partial sum. For instance, if we want the error to be less than 0.01 then it suffices to concentrate on a partial sum s_n such that $a_{n+1} < 0.01$. In this case we see that this works if only $\frac{1}{n+1} \leq 0.01$ or $n + 1 > 100 \Leftrightarrow n > 99$. So we may take s_{100} as estimate. Below we calculate the partial sum using Python.

```
from IPython.display import display, Math
import math

n = 100
s = sum(((-1)**k)/k for k in range(1, n+1))
S = -math.log(2)
err = abs(S - s)

display(Math(rf"s_{{{n}}} = {s:.10f},\qquad -\ln 2 = {S:.10f},\qquad |S-s_{{{n}}}| = {err:.2
```

$$s_{100} = -0.6881721793, \qquad -\ln 2 = -0.6931471806, \qquad |S - s_{100}| = 4.98e - 03.$$

9 Absolute convergence

We call $\sum_{n=1}^{\infty} a_n$ *absolutely convergent* if $\sum_{n=1}^{\infty} |a_n|$ converges.

Theorem 9.1 (Absolute convergence implies convergence). *If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.*

Proof

Consider the sequence $\{a_n\}_{n\in\mathbb{N}}$. Suppose it is absolutely convergent, and

$$\sum_{n=1}^{\infty} |a_n| = S.$$

Then $\sum_{n=1}^{\infty} 2a_n$ is convergent with sum $2S$. Now consider the sequence $\{b_n\}_{n\in\mathbb{N}}$ where $b_n = a_n + |a_n|$ for all n. We have that

$$0 \leq b_n \leq 2|a_n|.$$

For the partial sums we see that $\sum_{n=1}^{k} b_n$ is non-decreasing in k by nonnegativity of the elements b_n. Also $\sum_{n=1}^{k} b_n \leq 2S$. Then by the Monotonic Convergence Theorem we conclude that the partial sums of b_n must converge, and that $\sum_{n=1}^{\infty} b_n$ exists. Now realize that

$$\sum_{n=1}^{k} a_n = \sum_{n=1}^{k} b_k - \sum_{n=1}^{k} |a_n|$$

and that the right-hand partial sums both converge, so that the left-hand partial sums must also converge.

We cannot simply reverse the theorem. If a sequence is *convergent*, it does not mean that it is *absolutely convergent*. This is a very common misconception (!). See the example below.

Example 9.1 (convergent but not absolutely)**.** The alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

converges (alternating series test), but

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n}$$

diverges. So it is not absolutely convergent. Such series are called *conditionally convergent.*

10 Ratio test

For many sequences it may be hard to know whether the corresponding series converge. Fortunately, there are several criteria that can help. Here we focus on the **ratio test**, which shows that a series can often be compared to a geometric series—whose convergence behavior we understand well.[1]

Theorem 10.1 (Ratio test)**.** *Let $\{a_n\}$ be a sequence with $a_n \neq 0$ eventually.*

1. *If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L < 1$, then $\sum_{n=1}^{\infty} a_n$ converges (in fact absolutely).*
2. *If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L > 1$ or the limit is ∞, then $\sum_{n=1}^{\infty} a_n$ diverges.*

Proof

(i) Suppose

$$\left|\frac{a_{n+1}}{a_n}\right| \to L \quad \text{with } L < 1.$$

Then there exist a number $r \in (0, 1)$ and an index N such that for all $n \geq N$,

$$\left|\frac{a_{n+1}}{a_n}\right| < r.$$

Hence, iterating,

$$|a_{N+1}| < |a_N|\, r, \qquad |a_{N+2}| < |a_N|\, r^2, \qquad \dots, \qquad |a_{N+k}| < |a_N|\, r^k.$$

Therefore, for any $\ell \geq 1$,

$$\sum_{k=1}^{\ell} |a_{N+k}| \leq |a_N| \sum_{k=1}^{\ell} r^k.$$

Since $\sum_{k=1}^{\infty} r^k$ converges, the comparison test implies that $\sum_{k=1}^{\infty} |a_{N+k}|$ converges. Consequently,

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{N} |a_n| + \sum_{k=1}^{\infty} |a_{N+k}|$$

[1] There are other tests. One you may have discovered in earlier exercises is the **integral test**. Another is **D'Alembert's criterion**. For our purposes here, the ratio test suffices.

converges, so $\sum_{n=1}^{\infty} a_n$ converges absolutely (hence converges).
(ii) If

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L > 1 \quad \text{or} \quad \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \infty,$$

then for all sufficiently large n we have $|a_{n+1}| > |a_n|$. In particular, $|a_n|$ cannot tend to 0, so $a_n \nrightarrow 0$. By the term test for divergence (Theorem **??**), the series $\sum_{n=1}^{\infty} a_n$ cannot converge.

Some examples here!

11 Integral test

Theorem 11.1 (Integral test)**.** *Let $f : [1, \infty) \to \mathbb{R}_+$ be a continuous non-increasing function with the property that*

$$\lim_{x\to\infty} f(x) = 0.$$

Define the sequence $\{a_n\}_{n=1}^{\infty}$ by $a_n = f(n)$. Then:

(a) If $\displaystyle\int_1^{\infty} f(x)\,dx = \infty$, then $\displaystyle\sum_{n=1}^{\infty} a_n = \infty$.

(b) If $\displaystyle\int_1^{\infty} f(x)\,dx = L < \infty$, then $\displaystyle\sum_{n=1}^{\infty} a_n$ is convergent.

A proof is illustrated by the following two examples.

Example 11.1. Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}.$$

Then this is the series corresponding to $f(x) = \frac{1}{\sqrt{x}}$ which is decreasing and $\lim\limits_{x\to\infty} f(x) = 0$.

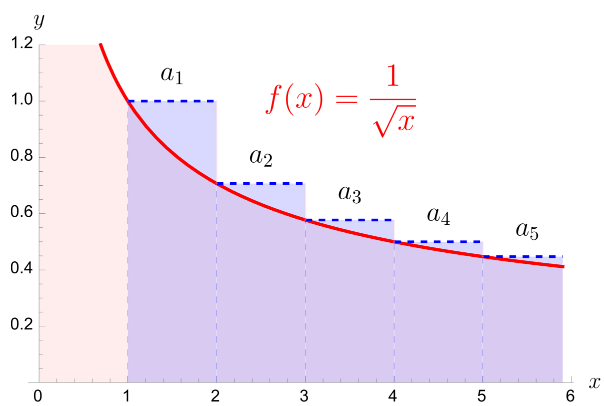

Figure 11.1: Partial sums s_n as blocks above the graph of f.

The total area of all the blocks is larger than the area between the graph of f and the x-axes. This area is given by

$$\int_1^\infty f(x)dx = \infty.$$

Therefore the series must diverge.

Example 11.2. Consider the series

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}.$$

Then this is the series corresponding to $f(x) = \frac{1}{x^2}$ which is decreasing and $\lim_{x\to\infty} f(x) = 0$.

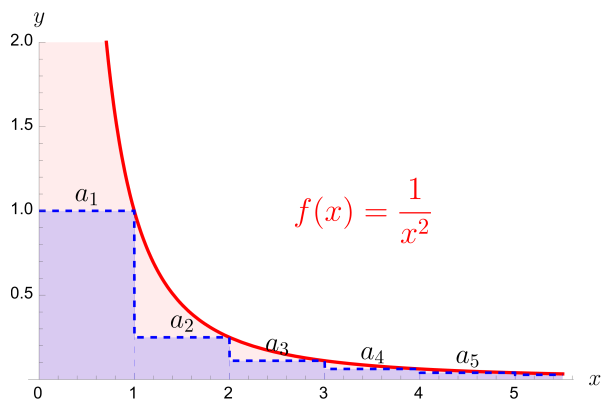

Figure 11.2: Partial sums s_n as blocks below the graph of f.

The total area of all the blocks is smaller than a_1 plus the area between the graph of f and the x-axes on $I = 1, \infty)$. This area is given by

$$\int_1^\infty f(x)dx = \left[-\frac{1}{x}\right]_1^\infty = 1 < \infty.$$

Therefore the sequence of partial sums is increasing and bounded, so it must converge.

You are invited to write down a general proof. The answer you can always find below.

Proof

(a) Let $k \in \mathbb{N}$ and $x \in [k, k+1]$. Since f is non-increasing, we have $a_k = f(k) \geq f(x)$, hence

$$a_k \geq \int_k^{k+1} f(x)\, dx.$$

Therefore,

$$\sum_{k=1}^{\infty} a_k \geq \int_1^{\infty} f(x)\,dx = \infty.$$

(b) Let $k \in \mathbb{N}$ and $x \in [k, k+1]$. Since f is non-increasing, we have $f(x) \geq f(k+1) = a_{k+1}$, hence

$$\int_k^{k+1} f(x)\,dx \geq a_{k+1}.$$

Summing from $k = 1$ to $n-1$ gives

$$\int_1^{n} f(x)\,dx \geq \sum_{k=2}^{n} a_k.$$

So for the partial sums $s_n = \sum_{k=1}^{n} a_k$,

$$s_n = a_1 + \sum_{k=2}^{n} a_k \leq a_1 + \int_1^{n} f(x)\,dx \leq a_1 + \int_1^{\infty} f(x)\,dx.$$

All terms a_k are nonnegative, so (s_n) is increasing, and the inequality above shows it is bounded. Hence (s_n) converges, so $\sum_{n=1}^{\infty} a_n$ converges.

A direct application is the following.

Theorem 11.2. *The series* $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^r}$ *is convergent if and only if* $r > 1$.

Proof

Try to prove it yourself. Calculate the integral $\displaystyle\int_1^{\infty} \frac{1}{x^r}\,dx$ and conclude.

Part III

Power series

12 Power series and convergence

12.1 Power series

Power series are special series, which take the form

$$\sum_{n=0}^{\infty} a_n x^n,$$

where $\{a_n\}_{n=0}^{\infty}$ is a sequence and $x \in \mathbb{R}$. We may wonder for what x values such series converge, or not.

As an example, consider the power series

$$\sum_{n=0}^{\infty} b_n := \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

where we took $a_n = \frac{1}{n!}$. A natural question here is for what x this sum makes sense. If we use the ratio criterion we get:

$$\left|\frac{b_{n+1}}{b_n}\right| = \left|\frac{\frac{x^{n+1}}{(n+1)!}}{\frac{x^n}{n!}}\right| = \left|\frac{x}{n+1}\right| \to 0, \quad \text{as } n \to \infty.$$

So we see that we have convergence for all $x \in \mathbb{R}$. In particular this means that the above power series as an infinite sum is always well-defined. So in this way we may define a function $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} x^n.$$

Later we will see that this is the well-known exponential function $f(x) = e^x$.

12.1.1 Convergence intervals

We will show that any power series $\displaystyle\sum_{n=0}^{\infty} a_n x^n$ converges on a symmetric interval around 0.

Theorem 12.1 (convergence interval)**.** *Consider a power series* $\displaystyle\sum_{n=0}^{\infty} c_n x^n$*. Then* *(i) If* $\sum_{n=0}^{\infty} c_n x^n$ *converges when* $x = b$*, then it converges when* $|x| < |b|$*.*

(ii) If $\sum_{n=0}^{\infty} c_n x^n$ *diverges when* $x = b$*, then it diverges when* $|x| > |b|$*.*

Proof

Case (i). Suppose that $\sum_{n=0}^{\infty} c_n b^n$ converges and $b \neq 0$. Then in particular

$$\lim_{n\to\infty} c_n b^n = 0.$$

So for $\varepsilon = 1$ there is a positive integer N such that

$$n \geq N \Rightarrow |c_n b^n| < 1.$$

Thus for $n \geq N$ we have

$$|c_n x^n| = \left|\frac{c_n b^n x^n}{b^n}\right| = |c_n b^n| \left|\frac{x}{b}\right|^n < \left|\frac{x}{b}\right|^n.$$

If $|x| < |b|$, then $|x/b| < 1$ so that $\sum_{n=0}^{\infty} |x/b|^n$ is a convergent geometric series. Then $\sum_{n=N}^{\infty} |c_n x^n|$ is also convergent. Hence the series $\sum_{n=0}^{\infty} |c_n x^n|$ is convergent (absolutely), and therefore $\sum_{n=0}^{\infty} c_n x^n$ is convergent.

Case (ii). Suppose that $\sum_{n=0}^{\infty} c_n d^n$ diverges. If x is any number with $|x| > |d|$, then $\sum_{n=0}^{\infty} c_n x^n$ cannot converge, because—by Case (i)—it would imply that $\sum_{n=0}^{\infty} c_n d^n$ converges.

This theorem shows that either there is a maximal $R \in \mathbb{R}_+$ with convergence on $I = (-R, R)$, or the interval of convergence is all of $\mathbb{R}$.

We cannot say so much about the end-points of those intervals; these have to be checked separately. The theory easily extends to cases where the power series *center* around a value $a \in \mathbb{R}$ instead of 0, i.e.

$$\sum_{n=0}^{\infty} c_n (x-a)^n$$

for a sequence $\{c_n\}_{n=0}^{\infty}$. We have the following.

Theorem 12.2 (convergence interval). *For a given power series* $\sum_{n=0}^{\infty} c_n (x-a)^n$*, there are only three possibilities:*

(i) The series converges only if $x = a$*.*

(ii) The series converges for all x*.*

(iii) There is a positive number R *such that the series converges if* $|x-a| < R$ *and diverges if* $|x-a| > R$*.*

Proof

First look at power series centered around $a = 0$: $\sum_{n=0}^{\infty} c_n x^n$.
Now suppose that for this series (i) and (ii) do not hold. Then there are nonzero numbers b and d such that $\sum_{n=0}^{\infty} c_n b^n$ converges and $\sum_{n=0}^{\infty} c_n d^n$ diverges. Therefore the set

$$S = \left\{ x : \sum_{n=0}^{\infty} c_n x^n \text{ converges} \right\}$$

is nonempty. We also know that the series diverges for any x with $|x| > |d|$, so S has an upper bound. Then S, being nonempty and bounded, has a least upper bound $R := \sup S$. If $|x| > R$, then $x \notin S$, so $\sum_{n=0}^{\infty} c_n x^n$ diverges. If $|x| < R$, then $|x|$ is not an upper bound for S, and so there is $b \in S$ with $b > |x|$. Since $b \in S$, $\sum_{n=0}^{\infty} c_n b^n$ converges, so must $\sum_{n=0}^{\infty} c_n x^n$.
Now consider the general case, with $a \neq 0$. Make the change of variable $u = x - a$. Then the power series becomes $\sum_{n=0}^{\infty} c_n u^n$ and we can apply the former argument to this series. In particular, in case (iii) we have convergence for $|u| < R$ and divergence for $|u| > R$. Thus we have convergence for $|x-a| < R$ and divergence for $|x-a| > R$.

The number R in Theorem Theorem 12.2 is called the *radius of convergence* of the power series. We can also summarize the three discussed cases by $R = 0$, $R > 0$, and $R = \infty$. If $R = 0$, the series converges only at a single point, whereas if $R = \infty$ convergence is established for all x.

12.2 Examples

Example 12.1 (exponential power series)**.** Consider the power series for the exponential function

$$\sum_{k=0}^{\infty} \frac{x^k}{k!},$$

and define $a_k = \frac{x^k}{k!}$. Then

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{\frac{x^{n+1}}{(n+1)!}}{\frac{x^n}{n!}}\right| = \frac{|x|}{n+1} \to 0 < 1.$$

So this series converges for **all** $x \in \mathbb{R}$.

Example 12.2. Define

$$f(x) = \sum_{k=1}^{\infty} \frac{x^k}{k}.$$

What is $\operatorname{dom}(f)$?

Use the ratio test with $a_n = \frac{x^n}{n}$. Then

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{\frac{x^{n+1}}{n+1}}{\frac{x^n}{n}}\right| = \frac{n}{n+1}|x| \to |x|.$$

So the series converges at least for $|x| < 1$, i.e. on $(-1, 1)$.
You can also check that it converges for $x = -1$, but not elsewhere. (Does this function look familiar?)

Example 12.3 (Binomial series)**.** The idea of binomials can be extended as follows. For $m \in \mathbb{R}$ and $k \in \mathbb{N}$ define

$$\binom{m}{k} := \frac{m(m-1)\cdots(m-k+1)}{k!}.$$

This agrees with the usual definition when $m, k \in \mathbb{N}$ and $k \leq m$:

$$\binom{m}{k} = \frac{m!}{k!(m-k)!}.$$

Consider $f(x) = (1+x)^m$ with $m \in \mathbb{R} \setminus \mathbb{N}$. Then the derivatives do not vanish. We compute

$$f^{(k)}(x) = m(m-1)\cdots(m-k+1)(1+x)^{m-k}.$$

The Taylor polynomial at $a = 0$ is

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k = \sum_{k=0}^{n} \binom{m}{k} x^k.$$

Now consider the limit (a power series):

$$g(x) = \lim_{n\to\infty} T_n(x) = \sum_{k=0}^{\infty} \binom{m}{k} x^k.$$

What is $\mathrm{dom}(g)$?

Apply the ratio test with $a_n = \binom{m}{n} x^n$:

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{\binom{m}{n+1} x^{n+1}}{\binom{m}{n} x^n}\right| = \left|\frac{m-n}{n+1} x\right| \to |x|.$$

So the series converges for $|x| < 1$ and diverges for $|x| > 1$. For $|x| < 1$ one obtains the classic identity

$$(1+x)^m = \sum_{k=0}^{\infty} \binom{m}{k} x^k.$$

Example 12.4. Find the Taylor series of $f(x) = (1+x)^{1/2}$.

Compute the first binomial coefficients:

$$\binom{1/2}{1} = \frac{1}{2}, \qquad \binom{1/2}{2} = \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!} = -\frac{1}{8}, \qquad \binom{1/2}{3} = \frac{\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)}{3!} = \frac{1}{16}.$$

Hence

$$\sqrt{1+x} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \cdots.$$

Example 12.5. Find the Taylor series of $f(x) = \sqrt{1-x}$.

Solution: substitute $-x$ for x in the previous example:

$$\sqrt{1-x} = 1 - \frac{1}{2}x - \frac{1}{8}x^2 - \frac{1}{16}x^3 - \cdots.$$

Part IV

Taylor polynomials

13 Polynomials and approximation

Polynomials are among the most important functions in mathematics and, more generally, in science. They are computationally convenient: function values and higher-order derivatives can be computed using only addition and multiplication—exactly the operations that lie at the heart of machine computation.

Polynomials are also used to approximate more complex functions. In *Mathematics 1* you saw how to approximate a function locally by its *linear approximation*—a polynomial of degree 1. Figure 13.1 illustrates this for $f(x) = e^x$, together with the linear approximations at $a = 0$ and $a = 1$.

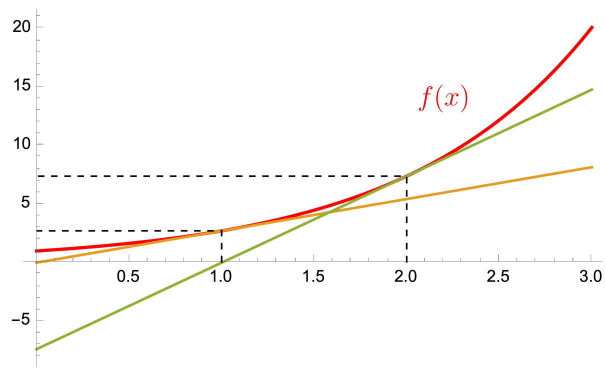

Figure 13.1: Plots of $f(x) = e^x$ together with the linear approximations at $a = 1$ and $a = 2$.

This chapter studies approximation by **Taylor polynomials**, which use information about local derivatives. In doing so, we will look at sequences of *functions* (Taylor polynomials of increasing degree) rather than sequences of numbers.

13.1 Road map

- **Tangency and Taylor polynomials** (how to match derivatives at a point)
- **Taylor's Theorem** (the remainder term and approximation quality)
- **Applications** (optimization, estimating e, the binomial formula)
- **Taylor series** (power series expansions)

14 Tangency and Taylor polynomials

Definition 14.1 (Tangency)**.** Let $f, g : \mathbb{R} \to \mathbb{R}$ be differentiable.

- The graphs of f and g **intersect** at $x = a$ if $f(a) = g(a)$.
- They are **tangent** at $x = a$ if

$$f(a) = g(a) \quad \text{and} \quad f'(a) = g'(a).$$

Example 14.1. Consider $f(x) = 2 - e^{x-1}$.

- The graph of $g_0(x) = 1$ merely intersects f at $x = 1$, because $f(1) = g_0(1) = 1$.
- The graph of $g_1(x) = 2 - x$ is tangent to f at $x = 1$, because $f(1) = g_1(1) = 1$ and $f'(1) = g_1'(1) = -1$.

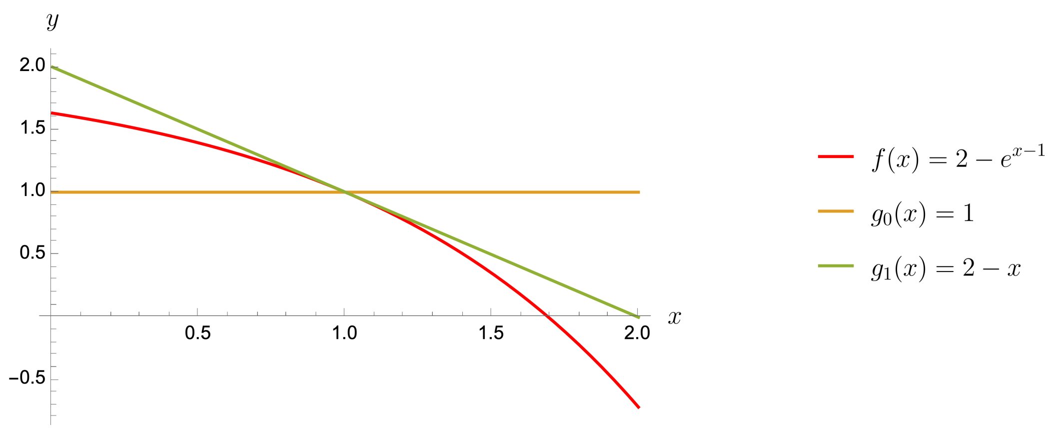

Figure 14.1: Tangency example.

14.1 Higher-order tangency

Two functions can be "more tangent" if more derivatives agree. We say the graphs of two n times differentiable functions f and g are **tangent at** a **to order** n if

$$f^{(k)}(a) = g^{(k)}(a) \qquad \text{for } k = 0, 1, \ldots, n.$$

In the example above, $g_2(x) = 1 - x - \frac{1}{2}x^2$ is tangent to f at $a = 1$ to order 2 (check it). Around the point of tangency, g_2 is typically a much better local approximation of f than g_1.

14.2 Taylor polynomials

Let $I \subset \mathbb{R}$ be a nonempty open interval and let $f : I \to \mathbb{R}$ have n continuous derivatives $f, f', f'', \dots, f^{(n)}$. Fix $a \in I$ and suppose we know

$$f(a), f'(a), f''(a), \dots, f^{(n)}(a).$$

We ask:

1. Does there exist a polynomial T of degree n whose derivatives match those of f at a up to order n?
2. Can we compute it efficiently?
3. How good is it as an approximation to f?

14.2.1 Deriving the coefficients

Start with a polynomial of degree n expanded around a:

$$p(x) = c_0 + c_1(x-a) + \cdots + c_n(x-a)^n = \sum_{k=0}^{n} c_k (x-a)^k.$$

Matching $p(a) = f(a)$ gives $c_0 = f(a)$. Differentiating,

$$p'(x) = \sum_{k=1}^{n} k c_k (x-a)^{k-1}, \quad \Rightarrow \quad p'(a) = f'(a) \implies c_1 = f'(a).$$

Differentiating again,

$$p''(x) = \sum_{k=2}^{n} k(k-1) c_k (x-a)^{k-2}, \quad \Rightarrow \quad p''(a) = f''(a) \implies c_2 = \tfrac{1}{2} f''(a).$$

Continuing in the same way, one finds that choosing

$$c_k = \frac{1}{k!} f^{(k)}(a) \qquad (k = 0, 1, \dots, n)$$

makes p tangent to f at a to order n.

Definition 14.2 (Taylor polynomial)**.** Let $I \subset \mathbb{R}$ be a nonempty open interval and let $f : I \to \mathbb{R}$ be n times continuously differentiable. For $a \in I$, the polynomial

$$T_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

is called the **Taylor polynomial of order n at a**.

14.3 Taylor polynomials of well-known functions

Example 14.2 (Exponential function)**.** For $f(x) = e^x$, we have $f^{(k)}(x) = e^x$ for all k, so $f^{(k)}(0) = 1$. Hence the Taylor polynomials at $a = 0$ are

$$T_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}.$$

Example 14.3 (Sine and cosine)**.** If $f(x) = \sin x$, then

$$f'(x) = \cos x, \quad f''(x) = -\sin x, \quad f^{(3)}(x) = -\cos x, \quad f^{(4)}(x) = \sin x,$$

so the derivatives repeat in a cycle. At $a = 0$ we obtain

$$f^{(2k+1)}(0) = (-1)^k, \qquad f^{(2k)}(0) = 0.$$

Therefore the Taylor polynomials of $\sin x$ at 0 are

$$T_{2n+1}(x) = \sum_{k=0}^{n} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!}.$$

Similarly, for $f(x) = \cos x$ we have

$$f^{(2k)}(0) = (-1)^k, \qquad f^{(2k+1)}(0) = 0,$$

so

$$T_{2n}(x) = \sum_{k=0}^{n} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!}.$$

15 Taylor's Theorem

Taylor polynomials match derivatives of a function at a point. The zero order Taylor polynomial agrees with the function value, the first order Taylor approximation agrees with function value and slope, the second order Taylorpolynomial does all this and agrees in curvature of the function. The natural next question is whether they also provide good approximations to the *function values* on some interval.

In *Mathematics 1* you saw that a linear approximation can be poor if curvature is large. The same idea appears here: approximation quality depends on higher derivatives.

Example 15.1 (bad linear approximation)**.** Consider

$$f(x) = x + 100x^2.$$

The Taylor polynomials around $a = 0$ are

$$\begin{aligned} T_0(x) &= f(0) = 0, \\ T_1(x) &= f(0) + f'(0)x = x, \\ T_2(x) &= f(0) + f'(0)x + \tfrac{1}{2}f''(0)x^2 = x + 100x^2. \end{aligned}$$

The remainder after the linear approximation is

$$R_1(x) = f(x) - T_1(x) = 100x^2,$$

which becomes large even for small deviations from 0.

15.1 The remainder term

The quality of the nth-order Taylor polynomial T_n is measured by the **remainder term**

$$R_n(x) := f(x) - T_n(x).$$

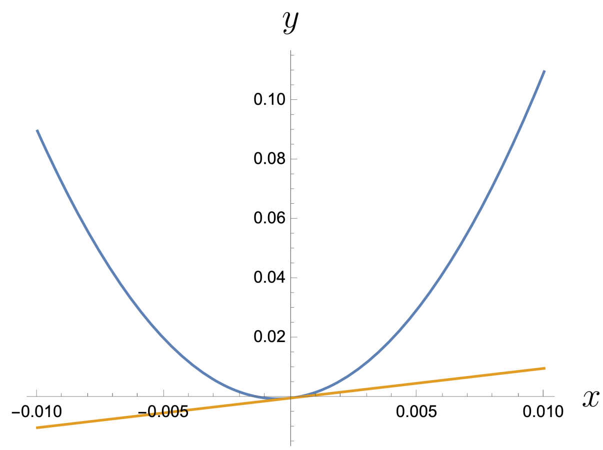

Figure 15.1: Here $T_2(x) = f(x)$, while the straight line $T_1(x)$ is a poor approximation.

15.2 Taylor's Theorem

Theorem 15.1 (Taylor)**.** *Let $f^{(n)}(t)$ be continuous for $a \le t \le x$ and let $f^{(n+1)}(t)$ exist for all $a < t < x$. Then there exists a point ξ_x between a and x such that*

$$R_n(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!}(x-a)^{n+1}.$$

If $x < a$, interpret "ξ_x between a and x" as $\xi_x \in (x, a)$.

To prove Taylor's Theorem, we first prepare two results.

Cauchy's Mean Value Theorem

Theorem 15.2. ***(Cauchy's Mean Value Theorem).***
Let $f, g : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Assume $g'(x) \neq 0$ for all $x \in (a, b)$. Then there exists $\xi \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}.$$

Proof.
The Mean Value Theorem implies there exists $\gamma \in (a, b)$ with

$$\frac{g(b) - g(a)}{b - a} = g'(\gamma) \neq 0,$$

so $g(b) \neq g(a)$. Define

$$\varphi(x) := f(x) + p\, g(x),$$

and choose p so that $\varphi(a) = \varphi(b)$:

$$f(a) + pg(a) = f(b) + pg(b) \quad \Longrightarrow \quad p = -\frac{f(b) - f(a)}{g(b) - g(a)}.$$

By Rolle's Theorem there exists $\xi \in (a, b)$ with $\varphi'(\xi) = 0$, i.e.

$$f'(\xi) + p\,g'(\xi) = 0 \quad \Longrightarrow \quad p = -\frac{f'(\xi)}{g'(\xi)}.$$

Equating the two expressions for p gives the result. □

Technical Lemma

Theorem 15.3. *Let I be an open interval and let $f : I \to \mathbb{R}$ be $(n+1)$ times differentiable on I, with*

$$f(a) = f'(a) = \cdots = f^{(n)}(a) = 0.$$

Then for any $x \in I \setminus \{a\}$ there exists a point ξ_x between a and x such that

$$f(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!}(x-a)^{n+1}.$$

Proof.
Assume without loss of generality that $a < x$, and let $g(x) = (x-a)^{n+1}$. Then

$$g(a) = g'(a) = \cdots = g^{(n)}(a) = 0, \qquad g^{(n+1)}(x) = (n+1)!.$$

Applying Cauchy's Mean Value Theorem repeatedly to (f, g), then (f', g'), and so on, yields points

$$x_1 \in (a, x),\ x_2 \in (a, x_1),\ \ldots,\ x_{n+1} \in (a, x_n)$$

such that

$$\frac{f(x)}{g(x)} = \frac{f'(x_1)}{g'(x_1)} = \cdots = \frac{f^{(n+1)}(x_{n+1})}{g^{(n+1)}(x_{n+1})} = \frac{f^{(n+1)}(x_{n+1})}{(n+1)!}.$$

Let $\xi_x := x_{n+1}$. Then

$$f(x) = \frac{f^{(n+1)}(\xi_x)}{(n+1)!}(x-a)^{n+1}.$$

□

Proof of Taylor's Theorem

Let $R_n(x) = f(x) - T_n(x)$. Since T_n matches derivatives of f at a up to order n, we have

$$R_n^{(k)}(a) = 0 \qquad (k = 0, 1, \dots, n).$$

Apply the previous theorem to R_n to obtain a point ξ_x between a and x with

$$R_n(x) = \frac{R_n^{(n+1)}(\xi_x)}{(n+1)!}(x-a)^{n+1}.$$

Because polynomial T_n is of order n, we have $T_n^{(n+1)}(x) \equiv 0$. So this means that $R_n^{(n+1)}(\xi_x) = f^{(n+1)}(\xi_x)$, which gives the desired formula. $\square$

15.3 Interpretation

Taylor's Theorem says that the approximation error near a depends on

- the distance $|x - a|$,
- the order n,
- and the behavior of the derivative $f^{(n+1)}$ between a and x.

If $f^{(n+1)}$ behaves regularly and x is close to a, the Taylor approximation is typically accurate; erratic higher derivatives can make approximation worse.

16 Applications

16.1 Taylor's theorem and optimization

Let $f : \mathbb{R} \to \mathbb{R}$ have continuous first and second derivatives. Suppose a is a critical point, i.e. $f'(a) = 0$. Then the first-order Taylor polynomial at a is constant:

$$T_1(x) = f(a) + f'(a)(x - a) = f(a).$$

Taylor's Theorem gives, for x near a,

$$f(x) = f(a) + \frac{f''(\xi_x)}{2}(x - a)^2$$

for some ξ_x between a and x.

- If $f''(a) > 0$, then for x close enough to a we also have $f''(\xi_x) > 0$, so $f(x) > f(a)$ for $x \neq a$ near a. Thus a is a **strict local minimum**.
- If $f''(a) < 0$, then $f''(\xi_x) < 0$ near a, so $f(x) < f(a)$ for $x \neq a$ near a. Thus a is a **strict local maximum**.

16.2 Estimating the number e

Using the Taylor expansion of e^x at $a = 0$,

$$e^x = \sum_{k=0}^{n} \frac{x^k}{k!} + \frac{e^\xi}{(n+1)!} x^{n+1}, \qquad (0 < \xi < x),$$

and setting $x = 1$, we obtain

$$e = \sum_{k=0}^{n} \frac{1}{k!} + \frac{e^\xi}{(n+1)!}.$$

Since $0 < \xi < 1$, we have $1 < e^\xi < e$, hence

$$\sum_{k=0}^{n} \frac{1}{k!} < e \leq \sum_{k=0}^{n} \frac{1}{k!} + \frac{e}{(n+1)!}.$$

Rearranging the upper bound gives

$$e < \frac{(n+1)!}{(n+1)!-1} \sum_{k=0}^{n} \frac{1}{k!}.$$

Define

$$a_n := \sum_{k=0}^{n} \frac{1}{k!}, \qquad b_n := \frac{(n+1)!}{(n+1)!-1} \sum_{k=0}^{n} \frac{1}{k!}.$$

Then $a_n < e < b_n$ and

$$0 < e - a_n < b_n - a_n = \frac{1}{(n+1)!-1} \sum_{k=0}^{n} \frac{1}{k!}.$$

Computing the values shows that $b_n - a_n < 0.01$ for $n = 5$:

n	a_n	$b_n - a_n$
1	2.000	2.000
2	2.500	0.500
3	2.666	0.1159
4	2.708	0.0228
5	2.717	0.0038

So $e \approx 2.717$ and the error is less than 0.004.

16.3 The binomial formula

An expression x^n is a *monomial* (from Greek *monos*, “single”). An expression $(x+a)^n$ is a *binomial*.

For small n:

$$\begin{aligned}
(x+a)^1 &= x + a, \\
(x+a)^2 &= x^2 + 2ax + a^2, \\
(x+a)^3 &= x^3 + 3x^2a + 3xa^2 + a^3.
\end{aligned}$$

In general:

Theorem 16.1 (Binomial formula). *For $x, a \in \mathbb{R}$ and $n \in \mathbb{N}$,*

$$\begin{aligned}(x+a)^n &= \sum_{k=0}^{n} \binom{n}{k} a^k x^{n-k} \\ &= \binom{n}{0} x^n + \binom{n}{1} x^{n-1} a + \cdots + \binom{n}{n} a^n,\end{aligned}$$

where

$$\binom{n}{k} = \frac{n!}{(n-k)!k!}.$$

The binomial coefficient $\binom{n}{k}$ counts the number of ways to select k objects from n when order does not matter—matching the idea that expanding $(x+a)^n$ corresponds to choosing a from exactly k of the n factors.

Proof via Taylor's Theorem

Let $f(x) = (x+a)^n$. Then

$$f^{(k)}(x) = \frac{n!}{(n-k)!}(x+a)^{n-k}, \qquad k = 0, 1, \dots, n,$$

and $f^{(n+1)}(x) = 0$ for all x.
At $x = 0$,

$$f^{(k)}(0) = \frac{n!}{(n-k)!} a^{n-k}.$$

The Taylor polynomial of order n at 0 is therefore

$$\begin{aligned}T_n(x) &= \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} a^{n-k} x^k \\ &= \sum_{k=0}^{n} \binom{n}{k} a^{n-k} x^k.\end{aligned}$$

Since $f^{(n+1)} \equiv 0$, Taylor's Theorem gives $R_n(x) = 0$ and hence $f(x) = T_n(x)$, which is exactly the binomial formula. □

17 Taylor series

17.1 From Taylor polynomials to series

Suppose $f : I \to \mathbb{R}$ is infinitely differentiable (all derivatives exist and are continuous). Then for each n we can form a Taylor polynomial T_n at a point $a \in I$, and write

$$f(x) = T_n(x) + R_n(x).$$

Now consider the sequence of polynomials

$$T_0, T_1, T_2, \dots$$

If for a given x we have

$$\lim_{n\to\infty} R_n(x) = 0,$$

then

$$f(x) = \lim_{n\to\infty} T_n(x),$$

and the limit may be representable as a **power series**.

17.2 Example: the exponential function

For $f(x) = e^x$ at $a = 0$,

$$T_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}.$$

Taylor's Theorem gives a remainder term of the form

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}\, x^{n+1} = \frac{e^{\xi}}{(n+1)!}\, x^{n+1}, \qquad \xi \in (0, x).$$

Assume $x > 0$. Then $0 < \xi < x$ implies $e^{\xi} \le e^x$, so

$$|R_n(x)| \le e^x \frac{|x|^{n+1}}{(n+1)!}.$$

To see that the right-hand side goes to 0 as $n \to \infty$, define $a_n := \frac{x^n}{n!}$. Then

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{|x|}{n+1} \to 0.$$

Hence $a_n \to 0$, and therefore $R_n(x) \to 0$. Consequently,

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

(You can check the case $x < 0$ similarly.)

17.3 Example: the geometric series

Consider $f(x) = \frac{1}{1-x}$ on $\mathbb{R} \setminus \{1\}$. One has the identity

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots = \sum_{n=0}^{\infty} x^n \qquad \text{for } x \in (-1, 1).$$

Compute derivatives:

$$f^{(k)}(x) = \frac{k!}{(1-x)^{k+1}}.$$

At $a = 0$,

$$f^{(k)}(0) = k!,$$

so the Taylor polynomial at 0 is

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k = \sum_{k=0}^{n} x^k = 1 + x + x^2 + \cdots + x^n.$$

On $(-1, 1)$ these Taylor polynomials converge to $f(x) = \frac{1}{1-x}$, but not on the full domain.

17.4 Example: sine and cosine series

For $f(x) = \sin x$ at $a = 0$,

$$T_{2n+1}(x) = \sum_{k=0}^{n} (-1)^k \frac{x^{2k+1}}{(2k+1)!}.$$

The remainder satisfies (using $|\sin| \le 1$ and $|\cos| \le 1$)

$$|R_n(x)| \le \frac{|x|^{n+1}}{(n+1)!} \to 0 \quad (n \to \infty),$$

so

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots.$$

Similarly,

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots.$$

Both series have radius of convergence $R = \infty$.